Introduction

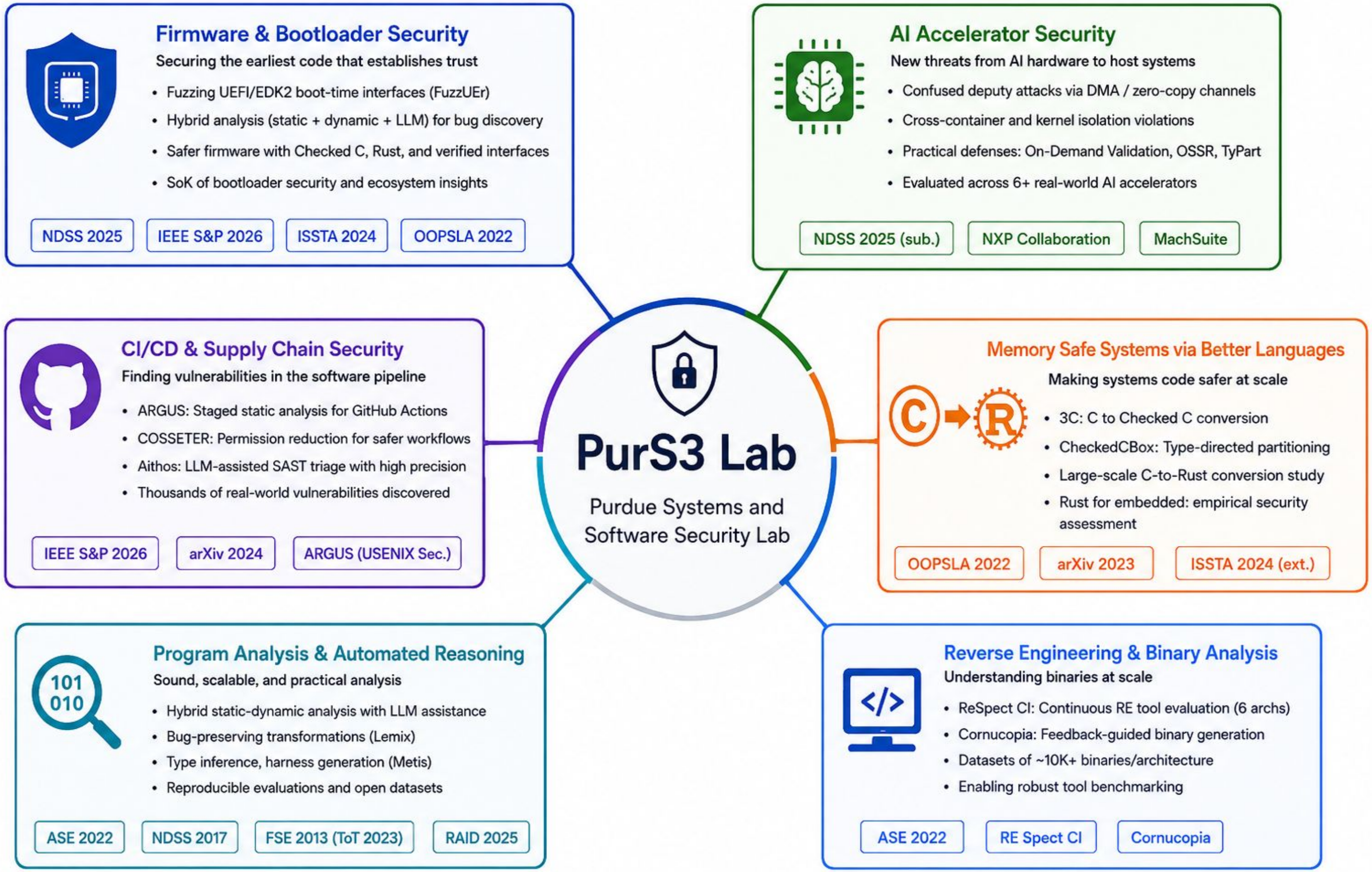
Hello!

I am Aravind Machiry

Assistant Professor

ECE

System and Software Security


Firmware & Bootloader Security
Securing the earliest code that establishes trust
Fuzzing UEFI/EDK2 boot-time interfaces (FuzzUEr)
Hybrid analysis (static + dynamic + LLM) for bug discovery
Safer firmware with Checked C, Rust, and verified interfaces
SoK of bootloader security and ecosystem insights
NDSS 2025
IEEE S&P 2026
ISSTA 2024
OOPSLA 2022
AI Accelerator Security
New threats from AI hardware to host systems
Confused deputy attacks via DMA / zero-copy channels
Cross-container and kernel isolation violations
Practical defenses: On-Demand Validation, OSSR, TyPart
Evaluated across 6+ real-world AI accelerators
NDSS 2025 (sub.)
NXP Collaboration
MachSuite
CI/CD & Supply Chain Security
Finding vulnerabilities in the software pipeline
ARGUS: Staged static analysis for GitHub Actions
COSSETER: Permission reduction for safer workflows
Aithos: LLM-assisted SAST triage with high precision
Thousands of real-world vulnerabilities discovered
IEEE S&P 2026
arXiv 2024
ARGUS (USENIX Sec.)
PurS3 Lab
Purdue Systems and Software Security Lab
Memory Safe Systems via Better Languages
Making systems code safer at scale
C
R
3C: C to Checked C conversion
CheckedCBox: Type-directed partitioning
Large-scale C-to-Rust conversion study
Rust for embedded: empirical security assessment
OOPSLA 2022
arXiv 2023
ISSTA 2024 (ext.)
Program Analysis & Automated Reasoning
Sound, scalable, and practical analysis
101
010
Hybrid static-dynamic analysis with LLM assistance
Bug-preserving transformations (Lemix)
Type inference, harness generation (Metis)
Reproducible evaluations and open datasets
ASE 2022
NDSS 2017
FSE 2013 (ToT 2023)
RAID 2025
Reverse Engineering & Binary Analysis
Understanding binaries at scale
</>
ReSpect CI: Continuous RE tool evaluation (6 archs)
Cornucopia: Feedback-guided binary generation
Datasets of ~10K+ binaries/architecture
Enabling robust tool benchmarking
ASE 2022
RE Spect CI
Cornucopia

What is IoT?

IoT Device: “Device that can interact with physical world and communicate with external entities.”

IoT System: A system of interconnected components that enable remotely interacting with physical world.

(Our definition)

Why we need IoT?

Automation, automation and automation!

- Ability to interact with physical world remotely.

IoT Systems: Characteristics and Requirements

- **Home Automation**: Physical override, Battery or AC Power. e.g., Door lock.
- Health care Automation: Size, Battery Powered, Resilient. e.g., Pacemaker.
- Connected and Autonomous Vehicles: Safe defaults, manual override, e.g., Waymo.
- Industry Automation: Reliability and Availability, Deterministic behavior, e.g., Smart Power grid.

IoT Building Blocks

- IoT Devices: Devices that directly interacts with physical world.

- Connectivity: A mechanism for the IoT device to communicate (mostly wirelessly) with external entities.

- Data Analysis (optional): An ability to collect/analyze data from the IoT device.

IoT Devices

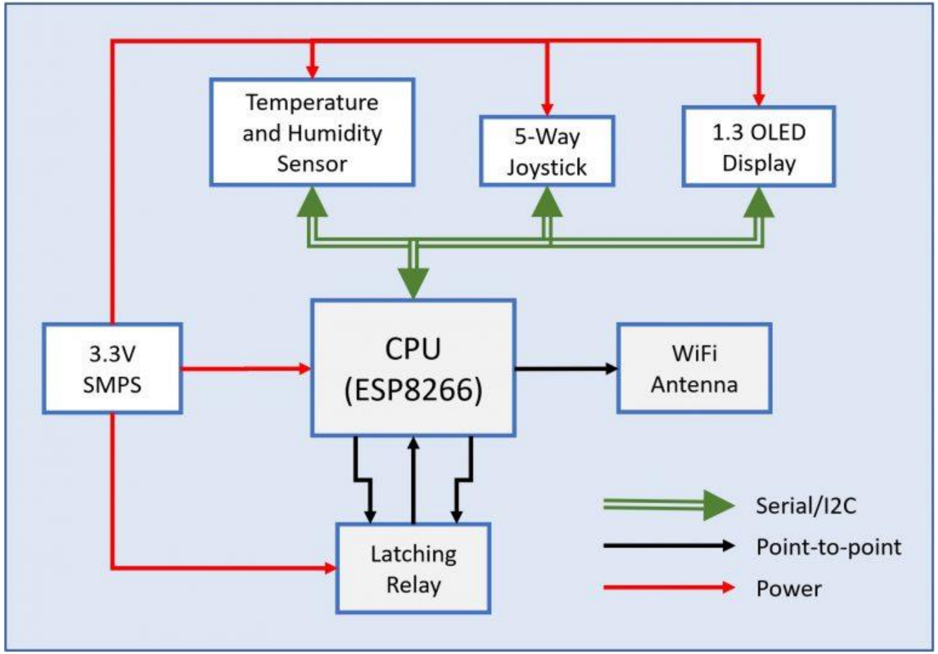
Interact with physical world and has connectivity!

- Sensors: Physical phenomenon to Digital data.
- Actuators: Digital command to physical action.
- Connectivity Mechanism: Bluetooth, WIFI, ZigBee, etc.

Typical IoT Device

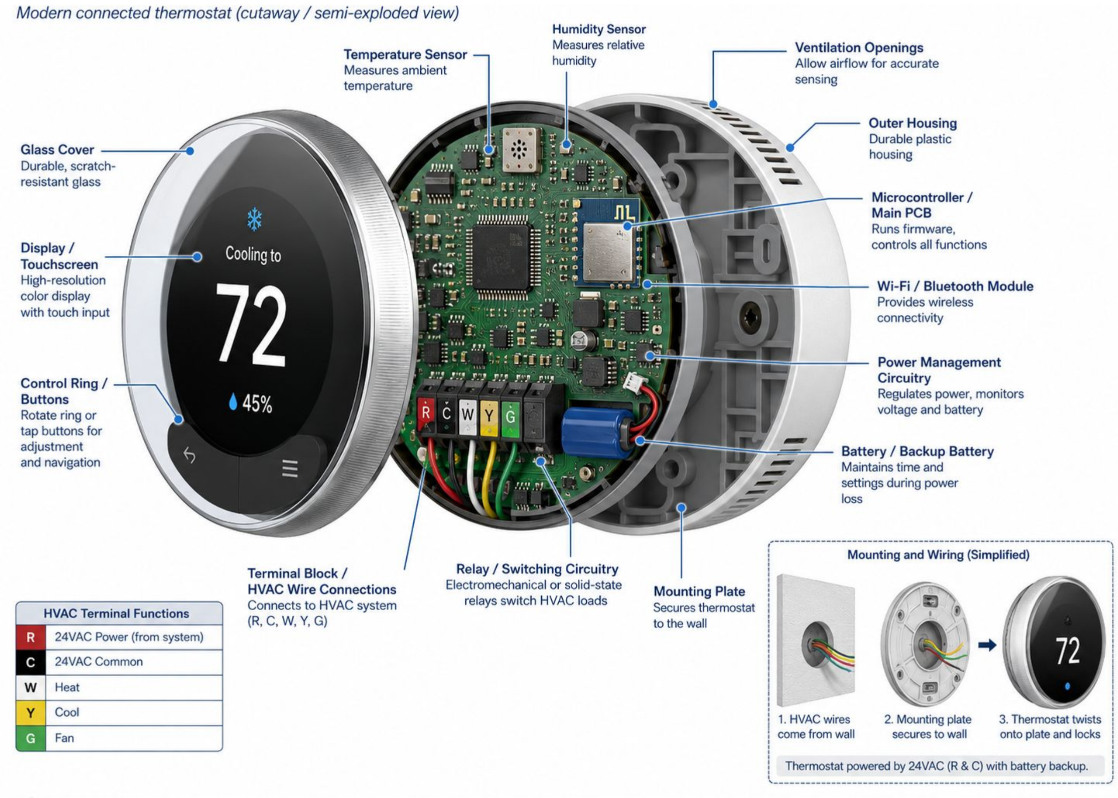
- Sensors and/or Actuators.
- Connectivity Mechanism: Bluetooth, WIFI, ZigBee, etc.
- Compute Unit: Processor to execute the application logic and interact with sensors/actuators and connectivity processor.

IoT Device (Thermostat)



https://addictek.com/en/category/iot/

IoT Device (Thermostat)



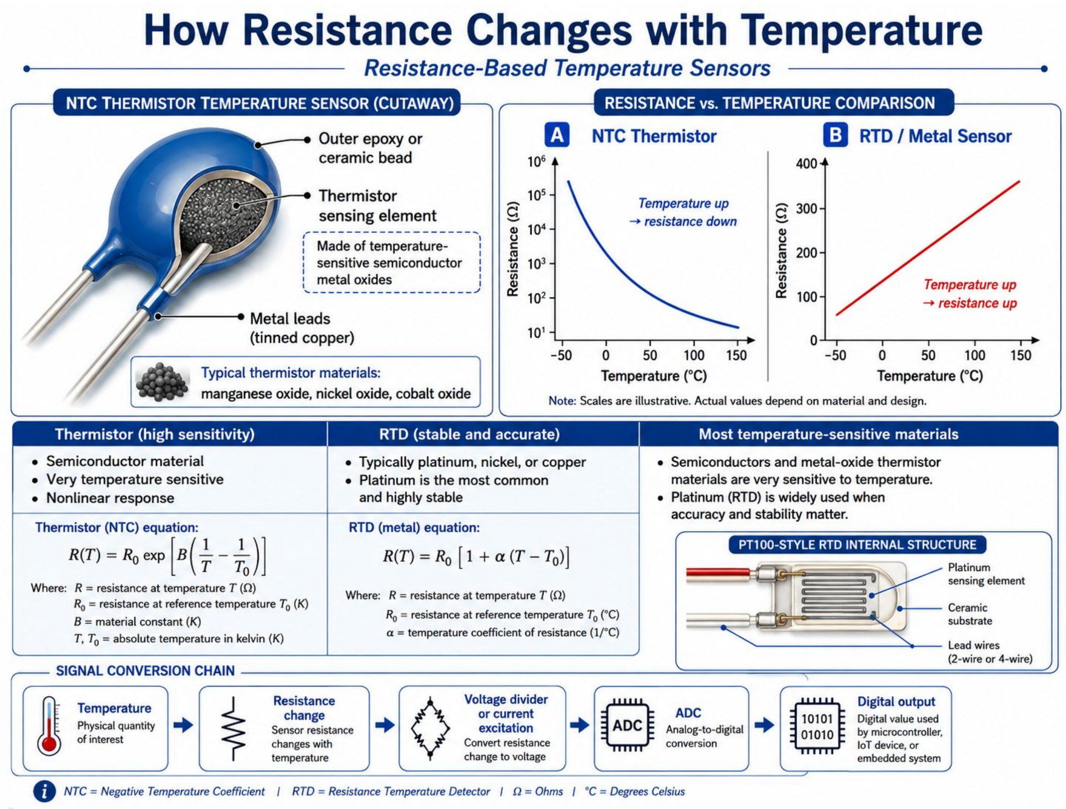
Common sensor chips

Sensor Principles

Materials are sensitive to certain physical attributes:

- Resistance of certain materials decreases with temperature. E.g., Nickel oxide.
- Thickness decreases capacity of Micro Electro Mechanical Systems (MEMS) spring.
- Flow increases the speed of MEMS fan.
  electromagnetic induction => change in electricity.

Sensor Principles



Connectivity

Various Technologies: Ethernet, WiFI, Bluetooth, BLE, ZigBee, etc.

How to choose?

Is there an ideal technology? I.e., works for all use cases.

Connectivity

Things to consider:

- Wired v/s Wireless.
- Range.
- Bandwidth.
- Power.

Connectivity

Things to consider:

- Wired v/s Wireless.
- Range: ~100 m (ZigBee) v/s ~500m (BLE) v/s Infinity (Ethernet/WIFI)
- Bandwidth: Kbps (ZigBee) v/s Mbps (BLE) v/s Gbps (WIFI)
- Power: Small battery (ZigBee/BLE) v/s rechargeable large battery (BLE/WIFI) v/s continuous power (WIFI)

Data Analysis

Usually performed through frameworks.

Examples:

Apache Spark: Distributed data processing framework.

- Average temperature in the house with (10 rooms and 10 sensors).
- Smoke diffusion in house through smoke detectors.

Grafana: Data visualization

- Temperature variance over the day across all rooms.

Definitions (in this class)

- IoT Device: (You know what this is)
- Controller: Entity that is used to communicate with IoT Device.
- Edge/Hub Device: Device (usually powered) that controls multiple IoT devices.
- Cloud Device: Device that enables communication with IoT Device or Hub via Internet.

Why do we need these?

- IoT Device.
- Controller.
- Edge/Hub Device.
- Cloud Device.

Why do we need these?

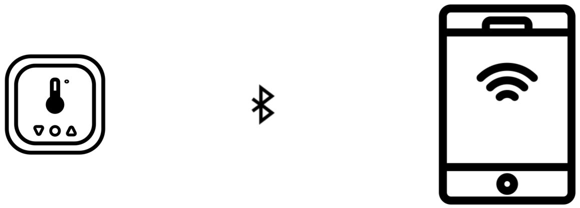
- IoT Device: To automate physical event.
- Controller: To easily communicate with the IoT device.
- Edge/Hub Device: To control multiple devices, have a common policy across devices, enable internet access to low range devices.
- Cloud Device: To interact with IoT Device over internet.

IoT System Architectures

- Direct Device Access: You access the device directly via controller.
- Hub Based Access: You access the device through Edge device/Hub.
- Cloud Based Access: You access the device through cloud (internet).

Direct Device Access

Controller accesses device directly.



Direct Device Access

Pros:

Cons:

Direct Device Access

Pros:

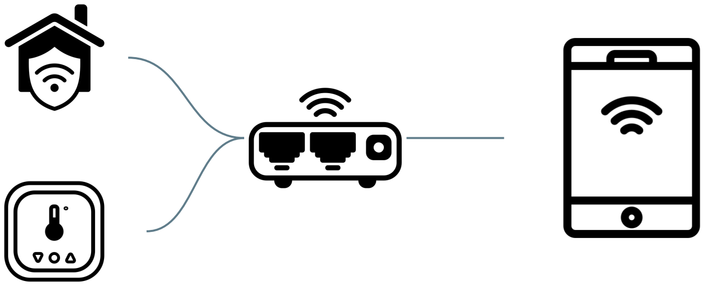
- Simple design.
- Low power requirement.
- Security through proximity.
- No need for external infrastructure.

Cons:

- Low range access.
- Limited scalability.

Hub Based Access

Controller accesses device through Hub.



Hub Based Access

Pros:

Cons:

Hub Based Access

Pros:

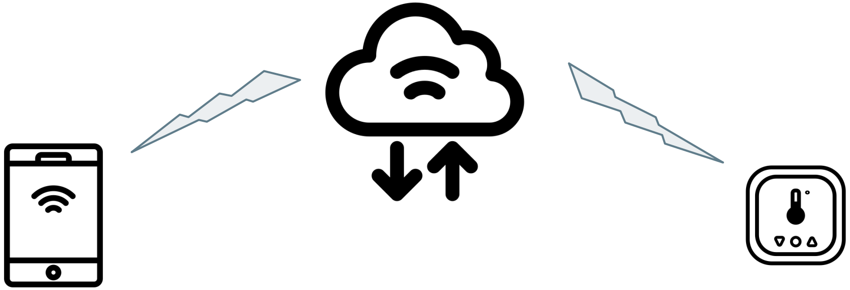
- High scalability.
- Uniform interface.
- Supports diverse IoT devices.

Cons:

- External infrastructure (Hub).
- Complex design.
- Low range access.

Cloud Based Access

Controller accesses device through cloud/internet.



Cloud Based Access

Pros:

Cons:

Cloud Based Access

Pros:

- Remote access.
- Uniform abstraction.

Cons:

- High power requirement.
- External infrastructure (internet).
- Complex software stack.

Best Architecture

Which is the best architecture?

Best Architecture

Low power IoT device: smart home lock

- Require proximity (for security).
- Battery powered.
- Need to be resilient.

Temperature sensors in hospital rooms.

- Multiple different types of sensors (ICU v/s regular room)

Door camera.

- Should be able to remotely see when someone is outdoor.

Logistics

- Web page: https://purs3lab.github.io/siot/
- The week's schedule will be updated on Sunday.
- TA: Sourag (Office hours: Tue/Thr: 11 am - 1 pm)

Grading

We will use the following distribution for grading:

- Four assignments (Total: 28%).
- Three Homeworks (30%).
- Midterm (20%).
- Final (20%).
- Class Participation (2%).

Class Style

- Half duplex.
- I expect everyone to participate.
- Ask questions.